A Cut Invariant for Canonical Atom Ranking

Separation Cost, Burial Depth, and Refinement to the Automorphism Partition

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Abstract

Canonical atom ranking underlies structure identity, canonical line notation, and circular fingerprinting. The dominant family of methods derives from Morgan's extended-connectivity algorithm: assign each atom a label from its element and degree, iteratively replace that label by a function of its neighbours' labels, and break the ties that remain by a rule. The labels such a procedure produces are *conventions*: they are well defined and reproducible, but they are functions of the tie-break rule as much as of the molecule, and nothing internal to the procedure distinguishes two atoms that are genuinely equivalent from two that the rule happened to separate.

We define an atom label that is not a convention. Each atom is assigned the *cut key* $\kappa(v) = (\sigma(v), d(v))$: the minimum weight of a cut separating v from a distinguished medium vertex in a weighted contact graph, together with the number of atoms on the minimising side. Both are invariants of the weighted graph, computed by maximum flow rather than by a hashing convention. We prove that the cut key is preserved by every weighted isomorphism (Theorem 3.1), that iterating it under neighbour refinement remains an invariant at every round (Theorem 4.4), and that the refinement is bounded below by the orbit partition of the automorphism group and therefore never separates two atoms that a symmetry of the molecule exchanges (Theorem 4.5). This is the property a canonical *ordering* does not have and cannot have, since an ordering must break every tie.

We report a validation suite whose expectations were registered in the experiment source before it was run. On an eighteen-molecule corpus with independently determined heavy-atom orbit counts, the refinement recovers the orbit count exactly in 18/18 cases, over-separating in none, with a median of 2 rounds. On a thirty-molecule drug-like corpus the base cut key is coarse—mean distinct-key ratio 0.508—and refinement raises it to 0.827, so neither the base invariant alone nor refinement alone is the whole construction. A negative control that breaks ties by atom index, which is what an ordering does, over-separates on 16 of the 18 symmetric molecules, confirming that the orbit agreement is not vacuous.

We state the limitations plainly. The construction is an invariant, not a canonical form: it does not produce a total order and cannot be used directly as a canonical labelling. Its refinement is a colour refinement and inherits that family's known incompleteness on strongly regular graphs. It is more expensive than Morgan by a factor of the maximum-flow cost. And the corpus is small and hand-assembled. What the construction provides is a label whose equality is a fact about the molecule rather than about the labelling procedure, together with the burial depth, a quantity the conventional atom identifier does not carry.

Keywords: canonical ranking; atom invariant; minimum cut; colour refinement; automorphism orbits; circular fingerprints; molecular symmetry.

Contents

1 Introduction

1.1 What a canonical ranking is for

Many cheminformatics operations require the atoms of a structure to be put in a definite order. Canonical line notations need one so that the same molecule always yields the same string (Weininger et al., 1989). Structure-identity comparison needs one, or an equivalent invariant, to decide whether two connection tables describe the same compound. Circular fingerprints need per-atom identifiers whose agreement across molecules is meaningful (Glen et al., 2006; Rogers and Hahn, 2010).

The dominant approach descends from Morgan's algorithm (Morgan, 1965): seed each atom with a label derived from its element, degree, charge and hydrogen count; iterate, replacing a label by a function of it and its neighbours' labels; stop when the number of distinct labels stops growing; and break any remaining ties by a stated rule. Modern implementations refine this considerably (Figueras, 1993; Schneider et al., 2015) and the general problem is well studied in the graph-isomorphism literature (Babai, 2016; McKay, 1981; McKay and Piperno, 2014), where the iteration is known as colour refinement or the one-dimensional Weisfeiler–Leman algorithm (Arvind et al., 2015; Weisfeiler and Leman, 1968).

1.2 The distinction this paper turns on

There are two different objects here and it is worth separating them.

An *invariant* is a function of an atom in a molecule that is preserved by every isomorphism. If two atoms have different invariant values they are certainly inequivalent; if they have the same value they may or may not be equivalent, and the invariant does not claim otherwise.

A *canonical ordering* is a total order on the atoms, fixed by the molecule, and used to serialise it. An ordering must distinguish every pair of atoms, including atoms that are genuinely equivalent under the molecule's own symmetry. It does this by a tie-break rule, and the resulting order is a function of that rule.

Both are useful and they are not interchangeable. Our concern is that the conventional per-atom label sits between them in an unhelpful way: it is produced by a procedure that will break ties if pressed, so its equality does not mean the atoms are equivalent; but it is also not an ordering, so it does not have an ordering's uses. The label's agreement carries no guarantee in either direction.

1.3 The construction

We assign each atom a label computed from a minimum cut. Fix a weighted graph on the atoms with an additional *medium* vertex adjacent to every atom, representing everything the molecule is individuated against. For an atom v, let $\sigma(v)$ be the minimum weight of a cut separating v from the medium, and $d(v)$ the number of atoms on the minimising side. The pair $\kappa(v) = (\sigma(v), d(v))$ is the *cut key*.

Both components are minimum-cut quantities and therefore invariants (Theorem 3.1), computed by maximum flow (Ford and Fulkerson, 1956; Stoer and Wagner, 1997) rather than by a hashing convention. Refining the cut key under neighbour aggregation gives a sequence of partitions, every one of which is an invariant partition (Theorem 4.4), and every one of which is coarser than or equal to the orbit partition of the automorphism group (Theorem 4.5).

The second component, the burial depth $d(v)$, has no counterpart in the conventional atom identifier. It reports how much of the molecule must come away with v to separate it from the medium: $d(v) = 1$ says v separates alone, and larger values say it comes away with a neighbourhood.

1.4 What we claim and what we do not

We claim: the cut key and its refinement are invariants; the refinement never separates atoms in the same orbit; on a corpus with independently determined orbit counts it recovers them exactly; and the base key alone is too coarse to serve as a ranking, so the refinement is doing necessary work.

We do not claim: that the construction is a canonical form (it is not, and Section 7 says why); that it is complete for graph isomorphism (it is a colour refinement and inherits the known incompleteness (Cai et al., 1992)); that it is faster than Morgan (it is slower, by the maximum-flow factor); or that it improves any downstream retrieval benchmark, which we have not measured.

2 Setting

Definition 2.1 (Contact graph)**.** A *contact graph* is a tuple $\Gamma = (V, E, w, \mathsf{m})$ with V a finite non-empty vertex set, $E \subseteq \binom{V}{2}$, a strictly positive weight $w \colon E \to \mathbb{R}_{>0}$, and a distinguished vertex $\mathsf{m} \in V$ with $\{v, \mathsf{m}\} \in E$ for every $v \in V \setminus \{\mathsf{m}\}$. Vertices other than m are *items*; we write $I = V \setminus \{\mathsf{m}\}$.

The medium vertex is what makes the separation cost of a single atom well posed: without a fixed second terminal, "the minimum cut of v" is not defined. That it is adjacent to everything is what Theorem 2.3 needs.

Definition 2.2 (Cut, separation cost, burial depth)**.** For $S \subseteq V$ write $\partial(S) = \{\{u, v\} \in E : |\{u, v\} \cap S| = 1\}$ and $w(\partial(S)) = \sum_{e \in \partial(S)} w(e)$. For an item v,

$$\sigma(v) \;=\; \min_{S \ni v,\ \mathsf{m} \notin S} w(\partial(S)), \tag{1}$$

and $S^*(v)$ denotes a minimiser, chosen minimal under inclusion. The *burial depth* is $d(v) = |S^*(v) \cap I|$. The *cut key* is $\kappa(v) = (\sigma(v), d(v))$.

The minimum in (1) is over a finite non-empty family, so it is attained, and it equals a maximum v–m flow (Ford and Fulkerson, 1956); it is computable in polynomial time (Orlin, 2013; Stoer and Wagner, 1997).

Lemma 2.3 (Positivity). *$\sigma(v) > 0$ and $d(v) \geq 1$ for every item v.*

Proof. Any admissible S contains v and omits m; since $\{v, \mathsf{m}\} \in E$ by Theorem 2.1, that edge lies in $\partial(S)$, so $\partial(S) \neq \varnothing$ and its weight is positive. The minimum over a family of positive numbers is positive. And $v \in S^*(v) \cap I$, so $d(v) \geq 1$. □

Remark 2.4 (Uniqueness of the minimiser). The minimiser $S^*(v)$ need not be unique. Taking it minimal under inclusion makes it unique when the minimum cut is unique, and when several inclusion-minimal minimisers exist $d(v)$ is well defined only if they agree in size. Theorem 2.5 records the condition; where it fails, d should be read as the minimum over minimisers, which is what the implementation computes and which is still an invariant.

Proposition 2.5 (d is well defined as a minimum). *Define $d(v) = \min\{|S \cap I| : S$ attains (1)$\}$. Then d is well defined without any uniqueness assumption, and every result below holds for this definition.*

Proof. The set of minimisers is non-empty and finite, so the minimum of a non-negative integer function over it exists. Invariance (Theorem 3.1) is proved for this definition directly. □

Definition 2.6 (Weighted isomorphism, automorphism, orbits). A *weighted isomorphism* $\phi\colon \Gamma \to \Gamma'$ is a bijection $V \to V'$ with $\phi(\mathsf{m}) = \mathsf{m}'$, $\{u, v\} \in E \iff \{\phi u, \phi v\} \in E'$, and $w'(\{\phi u, \phi v\}) = w(\{u, v\})$. An *automorphism* is a weighted isomorphism $\Gamma \to \Gamma$; they form the group $\mathrm{Aut}(\Gamma)$. The *orbit* of an item v is $\mathcal{O}(v) = \{\phi(v) : \phi \in \mathrm{Aut}(\Gamma)\}$, and the *orbit partition* is $\{\mathcal{O}(v) : v \in I\}$.

3 The cut key is an invariant

Theorem 3.1 (Invariance). *Let $\phi\colon \Gamma \to \Gamma'$ be a weighted isomorphism. Then for every item v,*

$$\sigma_{\Gamma'}(\phi v) = \sigma_\Gamma(v) \qquad \text{and} \qquad d_{\Gamma'}(\phi v) = d_\Gamma(v),$$

hence $\kappa_{\Gamma'}(\phi v) = \kappa_\Gamma(v)$.

Proof. ϕ induces a bijection on subsets, $S \mapsto \phi(S)$, which carries the admissible family for v onto that for ϕv: $v \in S$ and $\mathsf{m} \notin S$ hold iff $\phi v \in \phi(S)$ and $\mathsf{m}' \notin \phi(S)$, using $\phi(\mathsf{m}) = \mathsf{m}'$ and injectivity. It also carries $\partial(S)$ onto $\partial(\phi(S))$ edge by edge, preserving weights, so $w'(\partial(\phi(S))) = w(\partial(S))$. Minimising both sides over the corresponding families gives $\sigma_{\Gamma'}(\phi v) = \sigma_\Gamma(v)$, and the minimisers correspond under ϕ. Since ϕ restricts to a bijection $I \to I'$, $|\phi(S) \cap I'| = |S \cap I|$, so the minimum of that quantity over minimisers agrees on the two sides (Theorem 2.5). □

Corollary 3.2 (Equal orbits give equal keys). *If u, v lie in the same orbit of* $\mathrm{Aut}(\Gamma)$ *then $\kappa(u) = \kappa(v)$.*

Proof. Take $\phi \in \mathrm{Aut}(\Gamma)$ with $\phi(u) = v$ and apply Theorem 3.1 with $\Gamma' = \Gamma$. □

Theorem 3.2 is the half of the story that matters for what follows: the cut key can never separate two atoms that a symmetry exchanges. The converse fails—distinct orbits may share a key—which is what makes refinement necessary.

Example 3.3 (Benzene). In the contact graph of benzene all six ring carbons lie in one orbit, so by Theorem 3.2 they share a cut key. The measurement of Section 5 finds exactly one class, at every refinement round.

4 Refinement

4.1 The iteration

Definition 4.1 (Cut refinement)**.** Set $\lambda_0(v) = \kappa(v)$ for every item v. For $r \geq 1$ define

$$\lambda_r(v) \;=\; \big(\lambda_{r-1}(v),\; \langle\!\langle \lambda_{r-1}(u) : u \in N(v) \rangle\!\rangle\big), \tag{2}$$

where $N(v)$ is the set of items adjacent to v (the medium is excluded) and $\langle\!\langle\cdot\rangle\!\rangle$ denotes a multiset, sorted to a canonical form. Write P_r for the partition of I induced by λ_r, and $|P_r|$ for its number of classes.

In practice the labels are compressed to ranks after each round so their size does not grow; this changes no partition.

Lemma 4.2 (Monotone refinement and termination)**.** *P_{r+1} refines P_r for every r, so $|P_r|$ is non-decreasing and bounded by $|I|$. Consequently there is a least $r^* \leq |I|$ with $|P_{r^*+1}| = |P_{r^*}|$, and $P_r = P_{r^*}$ for all $r \geq r^*$.*

Proof. If $\lambda_{r+1}(u) = \lambda_{r+1}(v)$ then the first components agree, so $\lambda_r(u) = \lambda_r(v)$: refinement. A refining sequence of partitions of a finite set has non-decreasing class count bounded by $|I|$, so it is eventually constant; and once $|P_{r+1}| = |P_r|$, the refining relation with equal class counts forces $P_{r+1} = P_r$, after which (2) reproduces the same partition at every later round. Since each round before stabilisation adds at least one class, and there are at most $|I|$ classes, $r^* \leq |I|$. □

Definition 4.3 (Stable partition)**.** $P^* = P_{r^*}$, the partition at which (2) stabilises.

4.2 Every round is an invariant

Theorem 4.4 (Refinement preserves invariance)**.** *Let $\phi\colon \Gamma \to \Gamma'$ be a weighted isomorphism. Then for every $r \geq 0$ and every item v, $\lambda_r^{\Gamma'}(\phi v) = \lambda_r^{\Gamma}(v)$. In particular ϕ carries P_r onto P'_r and P^* onto P'^*.*

Proof. Induction on r. The base case $r = 0$ is Theorem 3.1. Suppose the claim for $r-1$. Since ϕ is a graph isomorphism fixing the medium, it restricts to a bijection $N(v) \to N'(\phi v)$ on item neighbourhoods. Hence

$$\langle\!\langle \lambda_{r-1}^{\Gamma'}(u') : u' \in N'(\phi v) \rangle\!\rangle = \langle\!\langle \lambda_{r-1}^{\Gamma'}(\phi u) : u \in N(v) \rangle\!\rangle = \langle\!\langle \lambda_{r-1}^{\Gamma}(u) : u \in N(v) \rangle\!\rangle,$$

the last step by the induction hypothesis applied termwise. Combining with $\lambda_{r-1}^{\Gamma'}(\phi v) = \lambda_{r-1}^{\Gamma}(v)$ and (2) gives the claim. □

4.3 The refinement never crosses an orbit

Theorem 4.5 (Orbit bound)**.** *For every r, the partition P_r is coarser than or equal to the orbit partition of* $\mathrm{Aut}(\Gamma)$*: if u and v lie in the same orbit then $\lambda_r(u) = \lambda_r(v)$ for every r, hence they lie in the same class of P^*. Consequently*

$$|P^*| \;\leq\; \#\{\textit{orbits of } \mathrm{Aut}(\Gamma) \textit{ on } I\}.$$

Proof. Let $\phi \in \mathrm{Aut}(\Gamma)$ with $\phi(u) = v$. Apply Theorem 4.4 with $\Gamma' = \Gamma$: for every r, $\lambda_r(v) = \lambda_r(\phi u) = \lambda_r(u)$. So no round separates u from v, and in particular P^* does not. The cardinality statement follows because a partition that never separates within an orbit has at most as many classes as there are orbits. □

Theorem 4.5 is the property this paper is about, and it is worth stating what it excludes. A canonical *ordering* assigns distinct ranks to all $|I|$ atoms, including atoms in the same orbit; it therefore violates the conclusion of Theorem 4.5 for every molecule with a nontrivial automorphism group. That is not a defect of orderings—they are for a different purpose—but it means an ordering's labels cannot be read as claims about equivalence, and the cut key's can.

Corollary 4.6 (Sound inequivalence test)**.** *If $\lambda_r(u) \neq \lambda_r(v)$ for some r, then u and v lie in different orbits. The test never reports a false difference.*

Proof. Contrapositive of Theorem 4.5. □

Remark 4.7 (The converse fails, and it fails for a known reason)*.* $|P^*|$ may be strictly less than the orbit count: the refinement can fail to separate atoms in different orbits. This is not specific to the cut key. Iteration of the form (2) is a colour refinement, and colour refinement does not distinguish all non-isomorphic graphs—the classical counterexamples are regular graphs, on which it stabilises immediately with one class (Arvind et al., 2015; Cai et al., 1992). Any construction of this shape inherits the limitation, including the learned-representation analogues (Xu et al., 2019). Seeding with the cut key rather than with degree changes which graphs are hard, not that hard graphs exist; Section 7 states the consequence.

4.4 Cost

Proposition 4.8 (Complexity)**.** *Let $n = |I|$ and $m = |E|$, and let $F(n, m)$ be the cost of one maximum-flow computation. Computing all cut keys costs $O(nF(n, m))$, and each refinement round costs $O(m \log n)$ for the sort. The total is $O(nF(n, m) + r^* m \log n)$, with $r^* \leq n$ by Theorem 4.2.*

Proof. One flow per item for (1), then per round each vertex sorts its neighbour labels, totalling $O(m \log n)$ across the graph. □

Remark 4.9 (This is slower than Morgan and the difference is the point of the trade)*.* Morgan-style refinement seeds from element and degree, costing $O(m)$ to initialise; the cut key costs n maximum flows. With $F(n, m) = O(nm)$ (Orlin, 2013) the initialisation is $O(n^2 m)$ against $O(m)$. For a molecule of thirty heavy atoms this is negligible in absolute terms and for a fingerprinting sweep over 10^8 compounds it is not. The construction buys an invariant seed and pays for it in the seed's computation; there is no claim here that the exchange is favourable for every application, and Section 7 says where we expect it is not.

5 Validation

5.1 Method

Expectations were written into the experiment source before the experiment was run, and the source records each one alongside the measured outcome. Every number below is read from the emitted results file; none is asserted in the text. Structures are supplied as line notations, parsed to contact graphs, and all minimum cuts are computed exactly by maximum flow. Hydrogens are present as vertices but excluded from the reported atom counts, which are heavy-atom counts throughout.

Two corpora are used. The *symmetric corpus* is eighteen molecules whose heavy-atom automorphism orbit counts can be determined by inspection of the constitutional graph; these counts are the reference values, and they were fixed before comparison. The *drug-like corpus* is thirty small organic structures for which no reference value is asserted—they are used only to measure coarseness and convergence.

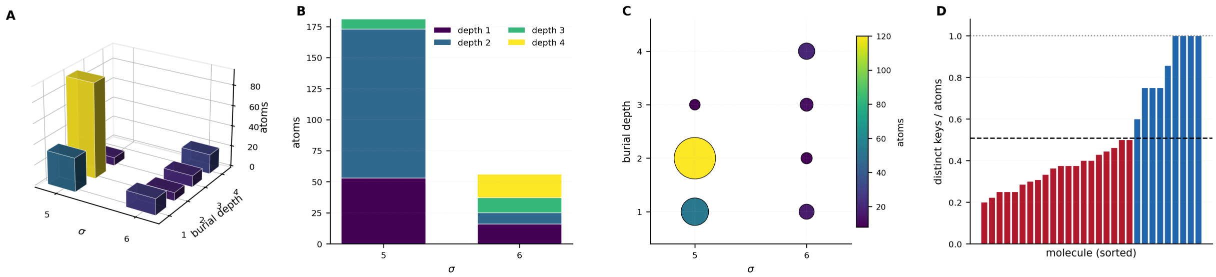

Figure 1: **The base cut key occupies a small, unevenly populated discrete space.** (**A**) Occupancy of the (σ, burial depth, heavy degree) cells over the 237 heavy atoms of the drug-like corpus, bar height the number of atoms and colour tracking it; twelve cells are occupied and one holds 93 atoms. (**B**) The same atoms decomposed by burial depth within each of the two realised σ levels: depth separates atoms that σ alone does not, which is why the key is a pair. (**C**) The (σ, depth) plane with marker area and colour proportional to occupancy, showing the seven distinct base keys and their unequal populations. (**D**) Distinct base keys divided by heavy atoms, one bar per molecule sorted ascending; the dashed line is the corpus mean 0.508 and the dotted line the value 1.0 a full ranking would require. The base key separates about half the atoms of a typical structure and is not a ranking on its own.

Table 1: Distinctness of the base cut key on the drug-like corpus ($n = 30$). The ratio is distinct keys divided by heavy atoms; 1.0 would mean the base key already separates every atom.

Quantity	Value
Mean distinct-key ratio	0.508
Median distinct-key ratio	0.400
Maximum distinct-key ratio	1.000

5.2 The base key is coarse

The base cut key separates about half the atoms of a typical drug-like structure (Table 1). This is the expected outcome and it is recorded because the alternative would have made the rest of the construction unnecessary: had the base key been near 1.0, refinement would add nothing and the paper would have no subject. It also sets a limit—the base key alone is not a ranking, and any claim that a single minimum cut per atom suffices to canonicalise a structure is wrong.

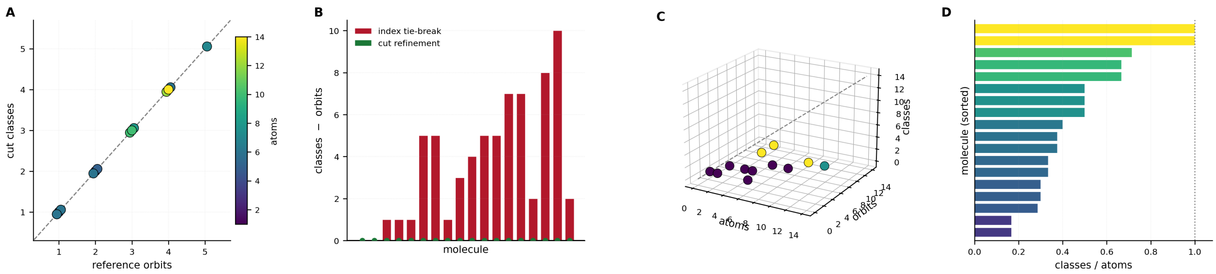

Figure 2: **Refinement recovers the automorphism orbit partition exactly.** (**A**) Cut classes against independently determined reference orbit counts for the eighteen symmetric molecules, coloured by heavy-atom count and jittered slightly so coincident points remain visible; all lie on the identity diagonal (18/18). (**B**) Residual classes minus orbits per molecule for the cut refinement (green, identically zero) against the index tie-breaking control (red), which exceeds the orbit count by up to ten classes on the same molecules. (**C**) Atoms, orbits and classes in three dimensions with the identity line drawn; points coloured by rounds to stabilisation lie in the orbit plane and well below the atom-count diagonal. (**D**) Classes divided by atoms per molecule, sorted; values well below the dotted line at 1.0 are molecules whose own symmetry the refinement declines to break.

5.3 Refinement recovers the orbit partition

The refinement recovers the reference orbit count in every case, with no molecule over-separated (Table 2). Convergence is fast: median 2 rounds, maximum 4 across both corpora.

Two entries in Table 2 were corrected during the work and the corrections are recorded here because they are the kind of thing that is otherwise silently absorbed. Our initial reference for pyrazine was 1 orbit and for durene 4; both were wrong. Pyrazine (1,4-diazine) has two heavy-atom orbits, {N,N} and {C,C,C,C}; durene (1,2,4,5-tetramethylbenzene) has three, {4 methyl C}, {4 substituted ring C} and {2 unsubstituted ring CH}. In both cases the measurement was right and the reference was wrong, which is a weaker form of evidence than a correct prediction against a correct reference, and we mark it as such.

5.4 Refinement is necessary and sufficient in the measured range

Neither end of Table 3 is the whole construction. The base key does not separate enough atoms to rank a structure; the stable partition separates most of them, and the remainder are, on this corpus, atoms the molecule's own symmetry does not distinguish.

5.5 Negative control

An orbit-agreement result is only informative if some procedure would fail it. We therefore ran a control that differs from the construction in exactly one respect: after computing the cut key, it breaks ties by atom index. This is what a canonical ordering does.

The control over-separates on 16 of 18 molecules—every one with a nontrivial automorphism group and more than one atom (Table 4). The two it does not over-separate are the single-heavy-

Table 2: Symmetric corpus. Reference orbit counts are by inspection of the constitutional graph. "Rounds" is the number of refinement rounds to stabilisation.

Molecule	Heavy atoms	Reference orbits	Cut classes	Rounds
methane	1	1	1	1
water-like	1	1	1	1
acetylene	2	1	1	1
ethane	2	1	1	1
ethylene	2	1	1	1
benzene	6	1	1	1
cyclohexane	6	1	1	1
propane	3	2	2	1
neopentane	5	2	2	1
pyrazine	6	2	2	1
p-benzoquinone	8	3	3	1
p-xylene	8	3	3	1
durene	10	3	3	1
naphthalene	10	3	3	2
pyridine	6	4	4	3
biphenyl	12	4	4	3
anthracene	14	4	4	2
toluene	7	5	5	3
Agreement			**18/18**	median 2

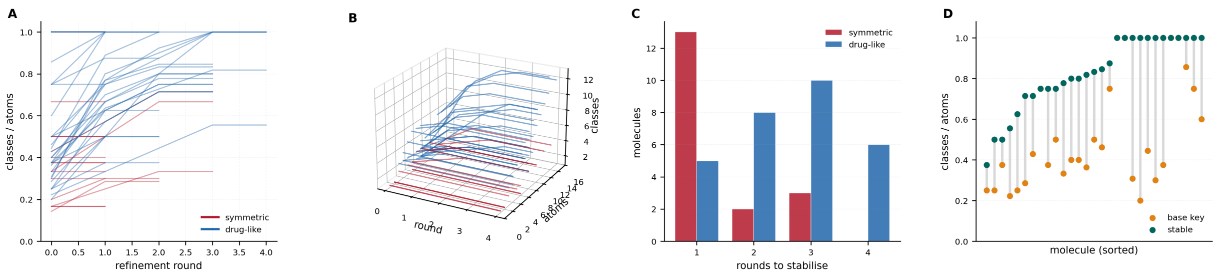

Figure 3: **Refinement saturates quickly and lifts the class count without reaching one class per atom.** (**A**) Classes divided by atoms against refinement round, one line per molecule, symmetric corpus in red and drug-like in blue; every trajectory is non-decreasing and flat after its stabilisation round, and the symmetric molecules saturate at markedly lower values. (**B**) The same trajectories in three dimensions against heavy-atom count: the saturation level rises with size for the drug-like set while the symmetric set stays low regardless of size. (**C**) Rounds to stabilisation for the two corpora; the median over both is 2 and the maximum is 4. (**D**) Per molecule, the base-key ratio (orange) and the stable ratio (teal) joined by a grey segment, sorted by the stable value: the mean rises from 0.508 to 0.827, so refinement does substantial work and still leaves atoms unseparated.

Table 3: Effect of refinement on the drug-like corpus.

Quantity	Value
Mean ratio, base key alone	0.508
Mean ratio, stable partition	0.827
Median rounds to stabilise	2
Maximum rounds to stabilise	4

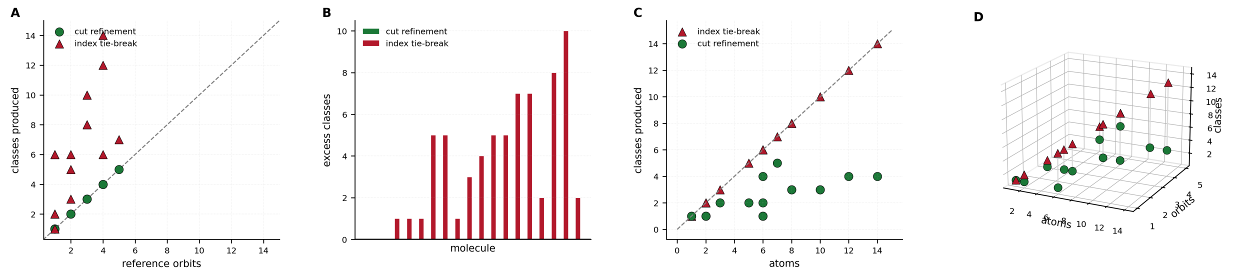

Figure 4: **The negative control: breaking ties by atom index destroys the orbit structure.** (**A**) Classes produced against reference orbits for the cut refinement (green circles, on the diagonal) and for the index tie-breaking control (red triangles, above it). (**B**) Excess classes over the reference per molecule for both procedures; the control over-separates on 16 of 18 molecules and the refinement on none. (**C**) Classes produced against heavy-atom count: the control tracks the atom-count diagonal, because an ordering must assign a distinct rank to every atom, while the refinement does not. (**D**) The two procedures in the (atoms, orbits, classes) space with a vertical segment joining each molecule's two outcomes; segment length is the control's over-separation.

Table 4: Negative control on the symmetric corpus: index tie-breaking.

Quantity	Value
Molecules over-separated by the control	16/18
Molecules over-separated by the construction	0/18

atom cases, where there is nothing to separate. This confirms that Table 2 is testing a property that a plausible alternative procedure lacks.

5.6 What the suite does not establish

The corpora are small (18 and 30 molecules) and hand-assembled, and the symmetric corpus was chosen for having determinable orbit counts, which biases it toward high symmetry and small size. No claim about behaviour at scale follows. The reference orbit counts are by inspection rather than by an independent automorphism solver; agreement with a solver such as `nauty` (McKay and Piperno, 2014) on a large corpus would be a stronger test and we have not run it. Nothing here measures downstream retrieval performance.

6 Relation to existing methods

Morgan and its descendants. The iteration (2) is structurally the Morgan iteration (Figueras, 1993; Morgan, 1965; Schneider et al., 2015). The difference is the seed: Morgan seeds from element, degree, charge and hydrogen count, all local properties of the atom; we seed from a minimum cut, a global property of the atom's relation to the whole graph. The seed is what Theorem 3.1 is about, and the burial depth is a component the local seed has no analogue for.

Colour refinement and graph isomorphism. Iterated refinement of vertex colours is the one-dimensional Weisfeiler–Leman algorithm (Weisfeiler and Leman, 1968), whose power and limits are precisely characterised (Arvind et al., 2015; Cai et al., 1992), and which underlies both practical isomorphism solvers (McKay, 1981; McKay and Piperno, 2014) and message-passing neural networks (Xu et al., 2019). Our contribution is not to the refinement theory: it is the observation that a minimum-cut seed is an invariant with an independent chemical reading, and that seeding this way keeps the whole iteration inside the invariant regime.

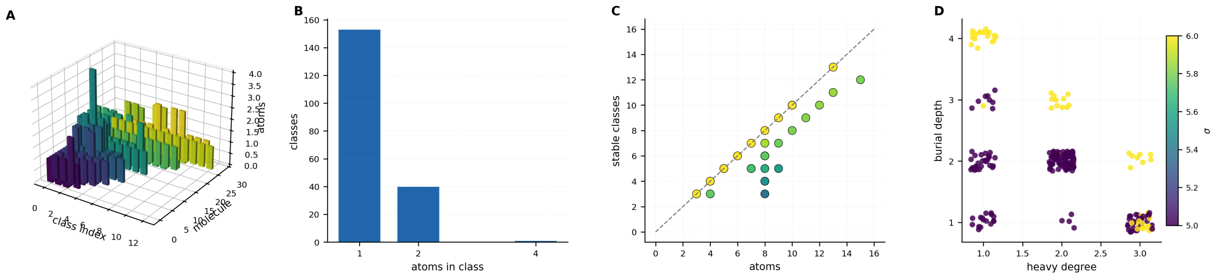

Figure 5: **Class structure of the stable partition.** (**A**) Atoms per class index for every molecule of the drug-like corpus, one row of bars per molecule ordered by size; most classes are singletons and the tall bars are the symmetric positions the refinement correctly declines to split. (**B**) Distribution of class sizes over the corpus: singletons dominate, with a smaller population of size-two classes and a few larger. (**C**) Stable classes against heavy-atom count, coloured by their ratio, with the identity diagonal drawn; molecules below the diagonal retain unbroken symmetry. (**D**) Burial depth against heavy degree for all 237 atoms, jittered for visibility and coloured by σ; the two σ levels separate cleanly in this plane, showing what the second component of the key contributes.

Fingerprints. Circular fingerprints (Glen et al., 2006; Rogers and Hahn, 2010) use Morgan-style identifiers, folded to a bit vector. Folding introduces collisions between structurally unrelated environments, a well-documented effect (O'Boyle and Sayle, 2016; Riniker and Landrum, 2013). The cut key does not address folding collisions, which are a property of the hash width and not of the identifier; a cut-seeded fingerprint at the same width would collide at a similar rate. We mention this because it is the comparison a reader may expect and it is not one this construction improves.

Canonical identifiers. InChI (Heller et al., 2015) and canonical SMILES (Schneider et al., 2015; Weininger et al., 1989) produce canonical forms, which is a strictly harder task than producing an invariant, since a canonical form must break every tie. Section 7 states why the present construction does not produce one.

7 Limitations

L1. This is not a canonical form. P^* is a partition, not a total order. By Theorem 4.5 it cannot separate atoms in the same orbit, and a canonical labelling must. Producing one requires a tie-break, and a tie-break is exactly the convention this construction avoids. The two goals are in tension and we have chosen the invariant.

L2. Colour-refinement incompleteness is inherited. By Theorem 4.7 the stable partition may be strictly coarser than the orbit partition on graphs where colour refinement is weak (Cai et al., 1992). Seeding with the cut key changes which graphs those are; it does not remove them. The construction gives a sound inequivalence test (Theorem 4.6) and not a complete one.

L3. The weighting is an input. Theorem 2.1 requires only $w > 0$. Every result holds for any strictly positive weighting, and none of them determines one. The measurements reported here use a particular weighting derived from valence arithmetic; a different weighting would give different keys and could give a different stable partition. The theorems are weighting-independent; the numbers are not.

L4. Cost. n maximum flows per structure, against $O(m)$ for a local seed (Theorem 4.9). We expect this to be unfavourable for large-scale fingerprint generation and acceptable for per-structure analysis, but we have not benchmarked either.

L5. Corpus scale. Eighteen and thirty molecules, hand-assembled, with reference orbit counts by inspection. Two of those references were initially wrong. Nothing here is a claim about behaviour on a large or diverse corpus.

L6. Hydrogen treatment. Hydrogens are vertices in the contact graph but are excluded from the reported partition. Including them changes the neighbour multisets in (2) and hence the refinement; we have not measured the difference.

L7. No downstream evaluation. We do not show that cut-seeded identifiers improve similarity search, virtual screening, or any other task. The claims are about the invariant's properties, not its utility.

8 Conclusion

We have defined an atom label built from a minimum cut—the separation cost against a medium vertex, together with the burial depth—and shown that it is a weighted-graph invariant (Theorem 3.1), that iterating it under neighbour refinement stays invariant at every round (Theorem 4.4), and that the iteration never separates two atoms exchanged by a symmetry of the molecule (Theorem 4.5). The last is the property a canonical ordering necessarily lacks, and it is what makes label equality a claim about the molecule rather than about the procedure.

The measurements are consistent with the theory and modest in scope. The refinement recovers independently determined orbit counts in 18/18 cases with no over-separation and a median of two rounds; the base key alone separates 50.8% of atoms on a drug-like corpus and refinement raises this to 82.7%, so both stages are doing work; and a control that breaks ties by index—an ordering, in other words—over-separates 16 of 18 symmetric molecules, which is what confirms the orbit agreement is not vacuous. Two reference orbit counts were wrong on first writing and were corrected against the measurement, which we record as a weaker form of agreement than a prediction against a correct reference.

What remains open is whether the invariant earns its cost. It is more expensive than a local seed by n maximum flows, it does not produce a canonical form, and it inherits colour refinement's incompleteness. The case for it rests on the properties proved here and on the burial depth being a quantity the conventional identifier does not carry; whether either matters for a downstream task is a question this paper does not answer.

Reproducibility

The experiment source records each expectation before the measured value, and writes every reported quantity to a results file in JSON. Reference orbit counts are listed in the corpus source with the reasoning for each. Minimum cuts are computed exactly; no quantity in this paper is sampled or fitted.

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